

Skin Lesion Analysis: Improving Early Detection through Multimodal Modeling

In dermatology, the distinction between a benign mole and a malignant melanoma is often invisible to the eye until it reaches a critical stage, making early intervention the most vital factor in patient survival. While computer vision models are adept at looking at lesions, clinicians never evaluate images in a vacuum. They instead weigh visual evidence against patient age, anatomical location, and clinical history to form a complete diagnostic picture.

You are being placed in the driver seat as a Junior Data Consultant to bridge the gap between computer vision and clinical reality using the MILK10k dataset. Your objective is to investigate the silent threat of misdiagnosis by comparing a standard visual-only tool against a multimodal model that integrates demographic data. This task requires navigating metadata cleaning and model training using timm library backbones to determine if this additional context improves sensitivity in catching true positives.

The importance of this work lies in its potential to catch aggressive cancers before they become life threatening. This project challenges you to move beyond simple accuracy scores and think critically about how model design impacts patient safety, specifically regarding the reduction of false negatives in oncology. By analyzing these results, you will provide actionable insights that could influence patient triaging and help healthcare providers prioritize urgent cases for immediate intervention.

By the end of this assignment, you will synthesize your findings into a Case Study Portfolio consisting of a reproducible GitHub repository and a strategic 10-slide presentation. The repository will house all cleaned data and scripts, while the presentation will serve as your primary method for communicating high-stakes findings to your peers. Your final deliverable must translate complex metrics like AUC into a clear narrative regarding real-world implications for dermatological practice.